

Large Language Model-assisted multi-scale hierarchical classification of ECG signals

Qianjiang Chen^a, Cheng Lian^a , Bingrong Xu^a, Quan Zhou^a, Yixin Su^a, Zhigang Zeng^b

^a School of Automation, Wuhan University of Technology, Wuhan, 430074, Hubei, China

^b School of Artificial Intelligence and Automation, Huazhong University of Science and Technology, Wuhan, 430074, Hubei, China

ARTICLE INFO

Keywords:

Hierarchical classification

Large language model

Multi-scale

Transformer

Temporal convolutional network

ABSTRACT

Deep learning models have been widely applied to electrocardiogram (ECG) classification. Existing research predominantly focuses on single-level classification tasks, while research on multi-level classification tasks remains relatively limited. Considering the hierarchical nature of heart diseases in the ECG dataset, we propose a hierarchical ECG classification model, which consists of two branches: Transformer and Temporal Convolutional Network (TCN). In the Transformer branch, the Fast Fourier Transform (FFT) is employed to extract periodic characteristics of the ECG signals, and the period length is selected as the multi-scale patch size to obtain multi-scale embedding vectors. Moreover, static patient information is transformed into prompt vectors using a pre-trained Large Language Model (LLM), which are integrated with embedding vectors to obtain more comprehensive representation. After integration, a router mechanism is also designed to learn the correlations of intra-scale and inter-scale features using self-attention. In the TCN branch, since ECG signals have complex and rapidly changing characteristics, convolution kernels of different scales are used to fully extract features. The feature vectors extracted from each level of the multi-scale Transformer and multi-scale TCN branches are fused accordingly, enabling the model to capture global and local information, thereby more fully understanding the potential patterns in the ECG signals. Extensive experiments are conducted on two different ECG datasets. The results demonstrate that the proposed model outperforms the competitive baselines in most metrics.

1. Introduction

Cardiovascular disease stands as one of the most formidable threats to human health globally. Electrocardiograms (ECGs) are important diagnostic tools for screening and evaluating cardiac abnormalities [1]. Healthcare institutions collect vast amounts of ECG data to analyze cardiovascular disease, significantly advancing the development of automated ECG analysis systems. However, several challenges remain, as collecting and analyzing critical ECG data require substantial financial resources and labor expenses.

In recent years, deep learning has made rapid advances in the field of computer vision. Techniques such as self-attention mechanisms [2, 3], cross-modal learning [4], and adversarial learning [5,6] have significantly improved feature extraction and data processing capabilities. In the domain of ECG classification, deep neural networks have been widely adopted for diagnosing ECG-related conditions due to their powerful information extraction abilities, and have increasingly become a prominent research focus. Several classic deep learning models

have been proposed for ECG signal classification, including multilayer perceptrons, convolutional neural networks (CNN), recurrent neural networks (RNN), long short-term memory (LSTM) networks, and Transformers. For instance, Zhang et al. [7] proposed a CNN-based method that can accurately classify a range of arrhythmias. In [8], an RNN ensemble with an attention mechanism was introduced to classify heart rhythms in ECG signals, achieving superior performance. Yan et al. [9] proposed a novel Transformer-based ECG classification method that achieved high classification accuracy.

However, existing models typically focus on classifying diseases at a single level, such as predicting several types of diseases at a specific level. Single-level classification models may encounter some issues in practical applications. For example, the relationships between different disease levels are not fully exploited. Heart diseases in the ECG dataset often exhibit a hierarchical structure, where certain fine-grained diseases are subsets of coarse categories. As shown in Fig. 1, the PTB-XL dataset [10] reveals a clear hierarchical structure in heart

* Correspondence to: School of Automation, Wuhan University of Technology, Wuhan 430074, China.

E-mail addresses: chenqianjiang@whut.edu.cn (Q. Chen), chenglian@whut.edu.cn (C. Lian), bingrongxu@whut.edu.cn (B. Xu), zhouquan910@whut.edu.cn (Q. Zhou), suyixin@whut.edu.cn (Y. Su), zgeng@hust.edu.cn (Z. Zeng).

<https://doi.org/10.1016/j.knosys.2025.113807>

Received 16 January 2025; Received in revised form 7 May 2025; Accepted 18 May 2025

Available online 3 June 2025

0950-7051/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

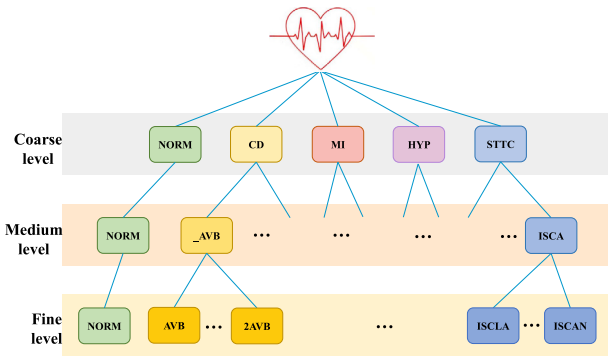


Fig. 1. The hierarchical heart disease structure in the PTB-XL dataset.

diseases, resembling a tree-like structure. Heart diseases are organized into three levels: coarse, medium, and fine, corresponding to 5, 23, and 44 categories, respectively. Ignoring the relationships between these levels may result in the model's inability to classify rare diseases accurately. In addition, cardiologists typically employ a hierarchical, step-by-step diagnostic approach when diagnosing cardiovascular diseases. In finer-grained diagnoses, the results of coarse-grained classification serve as critical reference information and constraints, enhancing both the accuracy and efficiency of the diagnostic process. This further confirms the importance of utilizing correlation information at different disease levels.

Moreover, most existing ECG classification models only utilize time-series data, overlooking static patient features like gender and age. [11] demonstrated that cardiovascular disease incidence is closely related to a patient's gender and age. As shown in Figs. 2 and 3, in the PTB-XL dataset, patients with different diseases exhibit significant variations in gender and age. Therefore, incorporating these static statistical information into the model could help improve model inference and enhance classification performance.

In addition, Large Language Models (LLMs) have made significant advancements in natural language processing and computer vision in recent years. As research deepens, these models are gradually being introduced to time series tasks, demonstrating strong feature modeling and generalization capabilities. However, most research focuses on using LLMs for time series prediction tasks currently, with fewer explorations into their application in time series classification.

Transformer and CNN models are widely used in time series classification tasks. Transformers, with their powerful global feature modeling capabilities and self-attention mechanism, are effective at capturing long-range dependencies and global contextual information. In contrast, CNN excels at capturing local patterns and correlations, with its localized receptive fields and efficient feature extraction capabilities. As a variant of CNN, Temporal Convolutional Networks (TCN) [12] not only retain the efficient characteristics of CNN but also enhance the modeling of time series features through innovations like causal convolutions and dilated convolutions. Given the complementary advantages of both approaches, we propose combining Transformers and TCN to build a classification model that can capture both global and local information, thereby enhancing the model's feature extraction capability and classification performance. In the field of computer vision, multi-scale feature learning plays a vital role in various visual tasks [13]. Tzimirou et al. [14] proposed that certain features in electroencephalography are closely related to different frequency bands. Moreover, multi-scale strategies are also widely used in ECG classification tasks. Jian et al. [15] proposed a CNN based on multi-scale feature concatenation for myocardial infarction detection from ECG signals. Their experimental results demonstrated that this approach outperforms single-scale methods. Similarly, Wang et al. [16] introduced a multi-scale fusion CNN that effectively captures abnormal disease patterns by extracting multi-scale features from ECG signals

and leveraging cross-scale information complementation. Inspired by this, we apply multi-scale techniques to both Transformers and TCNs, utilizing the feature vectors extracted at different scales for hierarchical ECG classification tasks.

In consideration of the above discussion, we propose a LLM-assisted multi-scale hierarchical ECG classification model in this paper. The model consists of two branches: the Transformer branch and the TCN branch. The Transformer branch utilizes periodic lengths derived from the Fast Fourier Transform (FFT) as the multi-scale patch lengths at coarse, medium, and fine levels. The patient's static information is transformed into prompts for a pre-trained LLM, generating prompt vectors fused with multi-scale patch embeddings. In addition, the model introduces routers in the attention layer, leveraging intra-scale and inter-scale self-attention to learn specific scale features and the correlations between different scales. In the TCN branch, the model utilizes three sets of different convolution kernels to extract local features, which are then used for ECG classification at three different levels. In the hierarchical structure, except for the first level, the output vectors of each level are fused with the associated information extracted from the previous level. Finally, the outputs from both branches at each level are fused to produce the final output vector for that level.

Our major contributions can be summarized as follows.

- We propose a hierarchical ECG classification model that fully utilizes the correlations between different levels, combining multi-scale Transformer and multi-scale TCN to merge global and local information.
- We generate prompts from the patient's static information and input them into a pre-trained LLM to assist in enhancing the model's classification performance.
- We introduce a router mechanism and use intra-scale and inter-scale attention to learn correlations both within the same scale and across different scales.
- Experimental results show that our model outperforms a series of CNN, RNN, and Transformer models in ECG classification tasks across all three levels.

The remainder of this paper is organized as follows: In Section 2, we review a series of deep learning models for ECG signal classification and the application of LLMs in time series tasks. Section 3 provides a detailed explanation of the proposed network. Section 4 outlines the experimental design. Section 5 presents the experimental results and analysis. Section 6 presents the visualization results. Section 7 discusses the limitations of the proposed model. Finally, Section 8 concludes the paper.

2. Related work

2.1. Deep learning models for ECG signal classification

Over the past few decades, the widespread application and success of deep learning methods in various fields have led researchers to propose several deep neural network architectures for ECG signal classification. Yıldırım et al. [17] proposed a CNN-based model for arrhythmia classification, which achieved high classification accuracy on the MIT-BIH arrhythmia database. Hou et al. [18] introduced an innovative deep learning network based on a LSTM autoencoder, which extracts features from arrhythmia data and uses a support vector machine to classify the arrhythmia based on the learned features. Natarajan et al. [19] integrated the Transformer into ECG signal classification, achieving outstanding performance.

CNN is particularly effective at extracting local features, demonstrating excellent generalization ability and faster convergence rates. In contrast, Transformers excel at capturing global semantic information and perform well on large datasets. Combining these two models could achieve better performance than using CNN or Transformer alone.

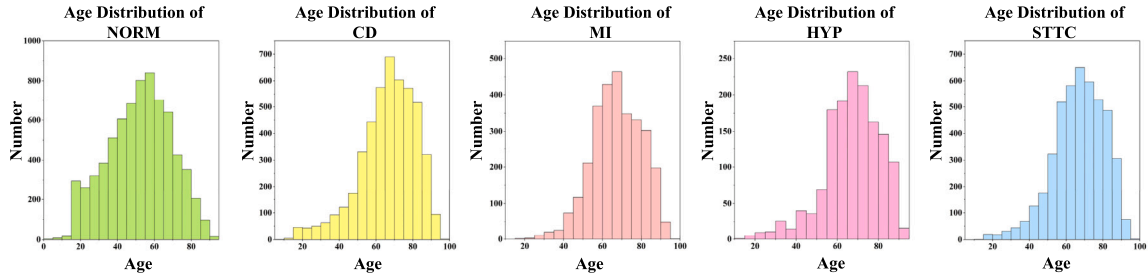


Fig. 2. Age distribution across the five super classes in the PTB-XL dataset.

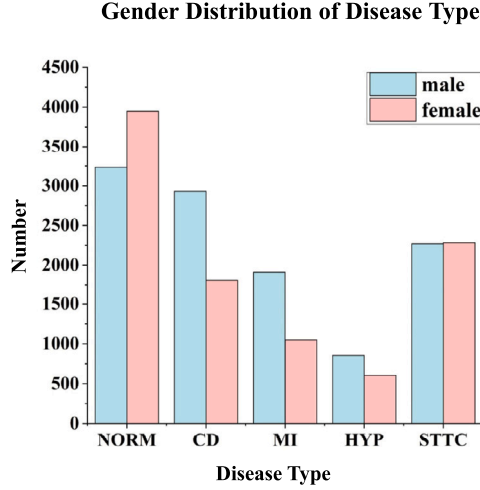


Fig. 3. Gender distribution across the five super classes in the PTB-XL dataset.

Yan et al. [20] proposed ConTNet, which combines Transformers and CNNs in a sequential architecture to provide a larger receptive field. Wu et al. [21] introduced a novel hybrid model called CTransCNN, which integrates CNN and Transformer models with an additional information interaction module, leading to improved model performance. However, the CNN-Transformer combined models mentioned above are mostly single-scale and perform only a single-level segmentation or classification task, which limits their potential. In our model, both CNN and Transformer are used in parallel, and both adopt a multi-scale strategy. Moreover, our multi-scale output is used for downstream hierarchical classification tasks. Compared to traditional models that perform a single task, our model can simultaneously handle three levels of classification tasks, significantly improving efficiency.

2.2. Applications of LLMs in time series analysis

In recent years, LLMs have garnered widespread attention in the fields of natural language processing and computer vision [22]. Inspired by their success, researchers are now exploring the potential of LLMs in time series analysis, with notable progress already achieved. For example, Jin et al. [23] proposed the TIME-LLM framework, which applies LLMs to time series forecasting through reprogramming, demonstrating exceptional performance in few-shot and zero-shot learning scenarios. Xue et al. [24] transformed numeric inputs and outputs into prompts, feeding them into a pre-trained LLM to make predictions in a sentence-to-sentence format. Cao et al. [25] introduced the TEMPO framework, which incorporates prompts and a novel seasonal trend decomposition within a pre-trained Transformer backbone, achieving outstanding predictive performance. However, most existing studies have focused on applying LLMs to time series forecasting tasks, with relatively fewer exploring their use in time series

classification tasks. In our model, we transform static information into prompts, which are input into a pre-trained LLM to generate prompt vectors that assist the model in completing the classification tasks.

3. Methods

3.1. Overall framework

The architecture of our model is illustrated in Fig. 4. We propose a novel dual-branch hierarchical ECG classification model that combines Transformer and TCN. The proposed model comprises pre-trained LLM, multi-scale embedding, encoder block, multi-scale TCN block, aggregator, fusion block, and classification block. In the Transformer branch, the model first divides the input 12-lead ECG signals into patches at three different scales: coarse, medium, and fine, with varying lengths. The patch lengths are determined based on the periodicity of the ECG signals, which is calculated through FFT. Furthermore, a pre-trained LLM is used to transform the patient's static information into prompt vectors, which are integrated with the multi-scale patch embedding vectors. Next, in the encoder block, we introduce routers that utilize the self-attention mechanism to learn the correlations of intra-scale and inter-scale features. The encoder produces outputs at three levels: coarse, medium, and fine, and each level of outputs are aggregated by an aggregator. In the TCN branch, the model utilizes three different sets of convolution kernels to extract local features, which are used for ECG classification at three distinct levels. In the hierarchical structure, the outputs of each level in both branches, except for the first level, incorporate the contextual information extracted from the previous level. Finally, a weighted fusion mechanism is applied to integrate the outputs of two branches at each level, producing the corresponding final output vector.

The multi-scale Transformer can simulate different frequency bands and capture global features, while the multi-scale TCN is capable of extracting local features from different ECG waveforms at various scales. By combining the multi-scale Transformer and multi-scale TCN for ECG classification, our model leverages the strengths of both models in capturing global dependencies and local features, enhancing the generalization capability.

3.2. Pre-trained LLM

Each patient's age and gender information is represented as discrete values. Based on the age and gender distribution of patients with heart disease mentioned above, the following prompt template is used: "This patient is {age} and {gender}". When designing the prompt, age information is categorized as follows: patients older than 60 are classified as elderly, those between 40 and 60 as middle-aged, and those younger than 40 as young. For gender information, a value of 1 indicates female, and 0 indicates male. The prompt is input to a pre-trained LLM and passed through a linear layer to generate a prompt vector $\mathbf{X}_{pro} \in \mathbb{R}^{1 \times d_m}$, where d_m represents the dimension of embedding. Considering our limited computing resources, there is only one RTX2080 Ti GPU. Therefore, the LLM chosen in this paper is BERT [26],

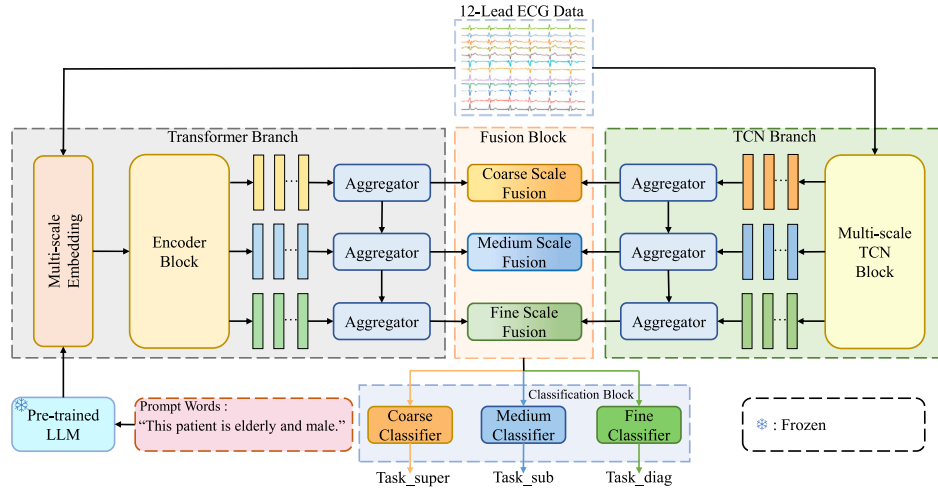


Fig. 4. Overview of our model. First, the ECG signals are processed through two branches for feature extraction, with each branch aggregating the extracted features at different levels. For each branch, except for the first level, the aggregated features at each subsequent level incorporate the aggregated features from the previous level. Next, the aggregated features from both branches are fused at different levels. Finally, three distinct classifiers are employed to perform hierarchical category predictions.

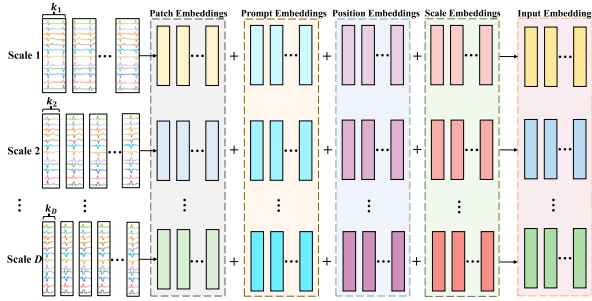


Fig. 5. Multi-scale embedding process. ECG signals are segmented into patches of varying lengths, which are embedded to obtain patch embeddings. These patch embeddings are fused with prompt embeddings, position embeddings, and scale embeddings to generate the final input embeddings.

which has smaller parameters than other larger models. Additionally, the BERT leverages a self-attention mechanism to learn contextual representations of text, making it well-suited for capturing relationships within the context. Naturally, in the case of rich computing resources, LLMs with larger number of parameters can be selected. By leveraging the pre-trained LLM, patient static information is integrated effectively into the main model, enhancing overall performance.

3.3. Multi-scale embedding

Considering an ECG signal $\mathbf{X} \in \mathbb{R}^{T \times C}$, where T represents the length, C represents the number of channels. The process of multi-scale embedding is shown in Fig. 5. A set $\mathbf{K} = \{K_1, K_2, \dots, K_D\}$ containing D patch sizes is defined, each corresponding to a patch segmentation operation.

In ECG signals, different frequency components correspond to distinct physiological features, such as heart rate. By applying the FFT, these frequency components can be effectively captured, enhancing the model's understanding of ECG signals. Regarding the selection of patch sizes, performing FFT online to extract the period of each sample would increase the model's complexity and computational load. Therefore, this paper adopts an offline approach, applying FFT to 500 randomly selected training samples. For each sample, the frequency components are ranked by amplitude, and the first S periods are selected. The average of these S periods across all samples is then computed. Subsequently, S is divided into three groups from large to small, and in each group,

$D/3$ period lengths are selected as patch sizes, resulting in a total of $D(D < S)$ patch sizes. These three groups of patches are applied to the subsequent three levels of ECG classification respectively.

For the patch size K_i corresponding to the i th scale, the input ECG signal is divided into N_i non overlapping patches $\mathbf{X}_p^i \in \mathbb{R}^{N_i \times K_i \times C}$. Zero padding is applied to ensure that the number of timestamps T is divisible by K_i , making $N_i = \lceil \frac{T}{K_i} \rceil$. Subsequently, a linear projection is used to map $\mathbf{X}_p^i$ to the latent embedding space and obtain the patch embedding $\mathbf{X}_{emb}^i \in \mathbb{R}^{N_i \times d_m}$, where d_m represents the dimension of embedding.

In addition, to leverage the powerful modeling capabilities of LLM to aid the model in performing classification tasks more effectively, we first generate prompt words for each sample based on its age and gender information. These prompts are then fed into a pre-trained LLM, which outputs prompt embedding $\mathbf{X}_{pro}^i \in \mathbb{R}^{1 \times d_m}$. The resulting embeddings are broadcast to produce $\tilde{\mathbf{X}}_{pro}^i \in \mathbb{R}^{N_i \times d_m}$.

Next, we incorporate position embeddings $\mathbf{E}_p \in \mathbb{R}^{H \times d_m}$, derived from positional encoding [27], where H is a large number. The first N_i rows of $\mathbf{E}_p$ are extracted to form the position embeddings $\mathbf{E}_{pos} = \mathbf{E}_p[1 : N_i] \in \mathbb{R}^{N_i \times d_m}$.

Then, to distinguish between different scales, we introduce a learnable scale embedding $\mathbf{X}_{sca}^i \in \mathbb{R}^{1 \times d_m}$ for each scale. This embedding is also broadcast to generate $\tilde{\mathbf{X}}_{sca}^i \in \mathbb{R}^{N_i \times d_m}$. Finally, the input embeddings for the i th scale is obtained as :

$$\mathbf{X}_{in}^i = \mathbf{X}_{emb}^i + \tilde{\mathbf{X}}_{pro}^i + \mathbf{E}_{pos} + \tilde{\mathbf{X}}_{sca}^i, \quad (1)$$

where $\mathbf{X}_{in}^i \in \mathbb{R}^{N_i \times d_m}$.

3.4. Encoder block

As shown in Fig. 6, the encoder block has L layers, each of which consists of an attention layer, two layers normalization layer, and a feedforward layers. In the attention layer, in order to relate information at different scales, this paper introduces a router for each scale:

$$\mathbf{r}^i = \mathbf{E}_p[N_i + 1] + \mathbf{X}_{sca}^i, \quad (2)$$

where $\mathbf{r}^i$, $\mathbf{E}_p[N_i + 1]$, $\mathbf{X}_{sca}^i \in \mathbb{R}^{1 \times d_m}$. The attention layer is then divided into intra-scale and inter-scale attention. It is worth noting that $\mathbf{E}_p[N_i + 1]$ is not used as positional embedding, but rather informs the router about the number of patches at the current scale, with the scale information included in $\mathbf{X}_{sca}^i$.

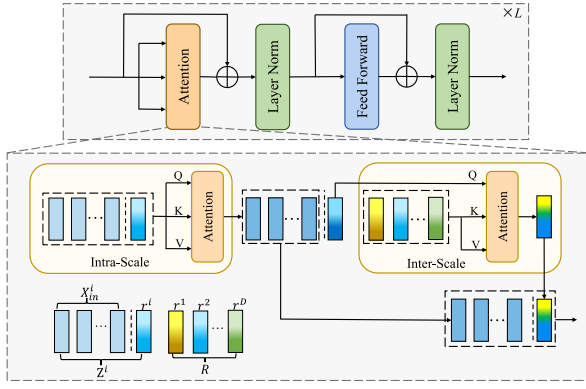


Fig. 6. The structure of encoder block.

At the intra-scale level, for the i th scale, we concatenate the input embedding X_{in}^i with the router vector r^i to form a new embedding Z^i :

$$Z^i = \text{Concat}(X_{in}^i, r^i), \quad (3)$$

where $Z^i \in \mathbb{R}^{(N_i+1) \times d_m}$, $\text{Concat}[\cdot, \cdot]$ denotes a concatenate operation. Then, self-attention is applied to Z^i :

$$Z^i \leftarrow \text{Attention}^{\text{Intra}}(Z^i, Z^i, Z^i), \quad (4)$$

where $\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V})$ represents the self-attention mechanism in [27]. Intra-scale self-attention allows the model to capture temporal features within a single scale, facilitating the extraction of local features and correlations between timestamps at the same scale. Meanwhile, the router vector is updated during intra-scale self-attention to summarize the features of that scale, thereby preparing for subsequent inter-scale attention.

At the inter-scale level, we concatenate the router embeddings $\{r^1, r^2, \dots, r^D\}$ to form a router embedding sequence $\mathbf{R}$:

$$\mathbf{R} = \text{Concat}(r^1, r^2, \dots, r^D), \quad (5)$$

where $\mathbf{R} \in \mathbb{R}^{D \times d_m}$, D denotes the number of scales. Next, for the i th scale, we perform self-attention on its router vector r^i and routing sequence $\mathbf{R}$:

$$r^i \leftarrow \text{Attention}^{\text{Inter}}(r^i, \mathbf{R}, \mathbf{R}). \quad (6)$$

Each router vector summarizes the global information of a specific scale during the intra-scale attention level. At the inter-scale attention level, by applying further self-attention between the router vectors, the model enables information exchange across different scales, effectively capturing features at each scale. After the inter-scale attention, the updated router vector r^i replaces the router vector in Z^i from the intra-scale attention level, resulting in the final output Z^i of the attention layer.

For the input embedding X_{in} , after passing through L layers of the encoder, we obtain a list of output vectors $\{Z^1, Z^2, \dots, Z^D\}$ arranged from the largest scale to the smallest scale. In the previous patch selection process, patches are divided into three levels: coarse, medium, and fine. Therefore, we consider dividing the output vector list into three parts: the first third is used for coarse-level classification, the middle third for medium-level classification, and the final third for fine-level classification.

3.5. Multi-scale TCN block

The multi-scale TCN block mainly consists of M layers of TCN residual blocks and a pooling layer, as shown in Fig. 7. Each TCN residual block primarily comprises two dilated causal convolution layers, two activation layers with ReLU functions, two weight normalization layers,

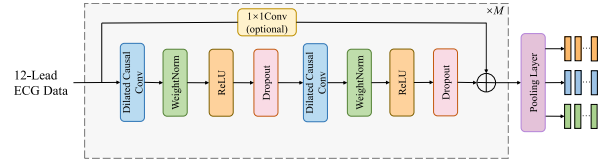
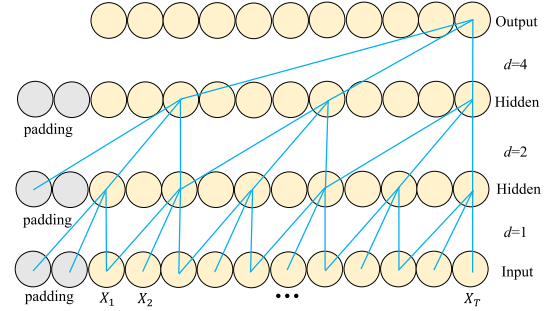


Fig. 7. The structure of Multi-scale TCN block.

Fig. 8. The structure of a dilated causal convolution with dilation factors $d = 1, 2, 4$ and convolution kernel size $f = 3$.

two dropout layers, and an optional 1D convolution. The shortcut connection uses an optimal 1×1 standard convolution to ensure that the output and input of the residual block have the same dimensionality. The internal structure of a dilated causal convolution layer with a convolution kernel size of $f = 3$ is shown in Fig. 8. Three different dilation factors ($d = 1, 2, 4$) are stacked in the convolution operations. From the bottom to the top, the receptive field size increases significantly. A larger receptive field allows the TCN to capture more historical information. Notably, the dilation factor increases progressively in the causal dilated convolution layers, but for the two dilated causal convolution layers within the same TCN residual block, both the dilation factor and kernel size are fixed.

For the input ECG signal $\mathbf{X} \in \mathbb{R}^{T \times C}$, a set $\mathbf{f} = \{f_1, f_2, \dots, f_n\}$ containing n convolution kernel sizes is defined. Then we divide them equally into three groups from large to small, with each group containing $n/3$ kernels. After passing through M layers of TCN residual blocks, we obtain three levels of output vector lists: $\{W_1, \dots, W_{n/3}\}$, $\{W_{n/3+1}, \dots, W_{2n/3}\}$, $\{W_{2n/3+1}, \dots, W_n\}$, where $W_i \in \mathbb{R}^{T \times d_m}$, $i = 1, 2, \dots, n$. The output vector corresponding to each convolution kernel undergoes an average pooling layer to obtain the final TCN branch output vector. Finally, we obtain three levels of the final output vector lists in the TCN branch: $\{W'_1, \dots, W'_{n/3}\}$, $\{W'_{n/3+1}, \dots, W'_{2n/3}\}$, $\{W'_{2n/3+1}, \dots, W'_n\}$, where $W'_i \in \mathbb{R}^{1 \times d_m}$, $i = 1, 2, \dots, n$. The final three output vector lists are used for coarse, medium, and fine level classification, respectively.

3.6. Aggregator

The Transformer and TCN branches produce coarse, medium, and fine outputs, which are aggregated by the aggregator block at the corresponding level. In the hierarchical structure, except for the first level, the output of each level contains contextual information extracted from the previous level.

In the Transformer branch, after passing through the encoder block, we obtain three levels of output vector lists: $\{Z^1, \dots, Z^{D/3}\}$, $\{Z^{D/3+1}, \dots, Z^{2D/3}\}$, $\{Z^{2D/3+1}, \dots, Z^D\}$. The aggregator consists of concatenation and average pooling operations.

For the coarse level, the output vectors in the coarse-level list are concatenated and averaged, resulting in the coarse-level output vector $\mathbf{Z}^{coarse}$ in the Transformer branch:

$$\mathbf{Z}^{coarse} = \text{Avgpool} \left(\text{Concat} \left(\mathbf{Z}^1, \dots, \mathbf{Z}^{\frac{D}{3}} \right) \right), \quad (7)$$

where $\mathbf{Z}^{coarse} \in \mathbb{R}^{1 \times d_m}$, $\text{Avgpool}[\cdot, \cdot]$ denotes an average pooling operation.

For the medium level, during the concatenation operation, the coarse-level output vector $\mathbf{Z}^{coarse}$ from the previous step is also included. After passing through the aggregator, we obtain the medium-level output vector $\mathbf{Z}^{medium}$ in the Transformer branch:

$$\mathbf{Z}^{medium} = \text{Avgpool} \left(\text{Concat} \left(\mathbf{Z}^{\frac{D}{3}+1}, \dots, \mathbf{Z}^{\frac{2D}{3}}, \mathbf{Z}^{coarse} \right) \right), \quad (8)$$

where $\mathbf{Z}^{medium} \in \mathbb{R}^{1 \times d_m}$. Since $\mathbf{Z}^{coarse}$ contains information from the coarse level, incorporating it into the medium-level output vector helps guide the medium-level classification task.

Similarly, for the fine level, the medium-level output vector $\mathbf{Z}^{medium}$ is also included during the concatenation operation. After passing through the aggregator, we obtain the fine-level output vector $\mathbf{Z}^{fine}$ in the Transformer branch:

$$\mathbf{Z}^{fine} = \text{Avgpool} \left(\text{Concat} \left(\mathbf{Z}^{\frac{2D}{3}+1}, \dots, \mathbf{Z}^D, \mathbf{Z}^{medium} \right) \right), \quad (9)$$

where $\mathbf{Z}^{fine} \in \mathbb{R}^{1 \times d_m}$. The medium-level information is incorporated into the fine-level output, aiding the fine-level classification task. Through hierarchical information propagation, the model generalization ability is enhanced, making the model more efficient and accurate for hierarchical ECG classification tasks.

In the TCN branch, for the three levels of the final output vector lists $\{\mathbf{W}'_1, \dots, \mathbf{W}'_{\frac{n}{3}}\}$, $\{\mathbf{W}'_{\frac{n}{3}+1}, \dots, \mathbf{W}'_{\frac{2n}{3}}\}$, $\{\mathbf{W}'_{\frac{2n}{3}+1}, \dots, \mathbf{W}'_n\}$ obtained from the multi-scale TCN block, the same concatenation and aggregation operation as in the Transformer branch are applied. The final output vectors at the three levels in the TCN branch are obtained as follows:

$$\mathbf{W}_{coarse} = \text{Avgpool} \left(\text{Concat} \left(\mathbf{W}'_1, \dots, \mathbf{W}'_{\frac{n}{3}} \right) \right), \quad (10)$$

$$\mathbf{W}_{medium} = \text{Avgpool} \left(\text{Concat} \left(\mathbf{W}'_{\frac{n}{3}+1}, \dots, \mathbf{W}'_{\frac{2n}{3}}, \mathbf{W}_{coarse} \right) \right), \quad (11)$$

$$\mathbf{W}_{fine} = \text{Avgpool} \left(\text{Concat} \left(\mathbf{W}'_{\frac{2n}{3}+1}, \dots, \mathbf{W}'_n, \mathbf{W}_{medium} \right) \right), \quad (12)$$

where $\mathbf{W}_{coarse}, \mathbf{W}_{medium}, \mathbf{W}_{fine} \in \mathbb{R}^{1 \times d_m}$.

3.7. Fusion block

After obtaining the output vectors at the three levels from both the Transformer and TCN branches, the corresponding level outputs from the two branches are fused and then input into their respective classifiers. For the fusion method, to reduce model complexity, a simple weighted fusion approach is employed, with the same weighting coefficient applied across all three levels:

$$\mathbf{X}_{total} = \alpha \mathbf{Z} + (1 - \alpha) \mathbf{W}, \quad (13)$$

where $\mathbf{X}_{total}, \mathbf{Z}, \mathbf{W} \in \mathbb{R}^{1 \times d_m}$, α is the hyperparameter. $\mathbf{X}_{total}$ represents the total output, while $\mathbf{Z}$ and $\mathbf{W}$ denote the outputs from the Transformer and TCN branches, respectively, at the same level.

3.8. Loss

The output vectors from the three levels are input into the coarse, medium, and fine-level classifiers to perform the downstream tasks: task_super, task_sub, and task_diag, corresponding to the three classification tasks. The overall loss of the model is the weighted sum of the losses from the three tasks:

$$\text{Loss}_{total} = \lambda \text{Loss}_{task_super} + \eta \text{Loss}_{task_sub} + \mu \text{Loss}_{task_diag}, \quad (14)$$

Table 1

Number of ECG records in the training, validation, and test sets for different tasks in the PTB-XL and SPH dataset.

Dataset	Task	Classes	Train	Val	Test	Total
PTB-XL	super	5	13 746	1718	1719	17 183
	sub	23	13 746	1718	1719	17 183
	diag	44	13 746	1718	1719	17 183
SPH	super	2	15 073	1884	1885	18 842
	sub	11	15 073	1884	1885	18 842
	diag	42	15 073	1884	1885	18 842

where Loss_{task_super} , Loss_{task_sub} , Loss_{task_diag} represent the losses for the three classification tasks respectively. λ , η and μ are hyperparameters. Since our hierarchical model relies on information from the previous level for guidance, it is crucial to ensure the accuracy of the higher-level classification while maximizing the accuracy of the lower-level tasks. In real-world cardiology diagnostics, the initial coarse assessment serves as a crucial reference for subsequent detailed classification. An error at this stage can have adverse consequences for further specialization. Therefore, in the loss function design, the weight for coarse classification is the biggest, followed by medium-level classification, with fine-grained classification having the smallest weight. Based on empirical experience, we set the weights as $\lambda = 0.6$, $\eta = 0.3$, and $\mu = 0.1$.

4. Experimental design

4.1. Datasets and preprocessing

The PTB-XL dataset [10] is a publicly available 12-lead multi-label ECG dataset. It contains 21,837 10-s clinical ECG recordings from 18,885 patients. The dataset also includes metadata such as age and gender, although some of the metadata is missing. The dataset comprises a total of 71 unique ECG statements, which are categorized into 44 diagnostic statements, 19 formal statements, and 12 rhythm statements. Notably, 4 formal statements and 4 rhythm statements overlap. According to the hierarchical structure, the 44 diagnostic statements are further divided into 23 subclasses and 5 coarse superclasses. In this paper, we perform hierarchical multi-task classification for the diagnostic statements (task_diag), subclasses (task_sub), and superclasses (task_super). Due to the necessity of using age and gender information for the model, we exclude samples with missing values for these attributes. In this paper, the sampling frequency of ECG is 100 Hz. The dataset is relatively clean, requiring only normalization, with no additional preprocessing steps needed.

The SPH dataset [28] is a publicly accessible 12-lead, multi-label ECG dataset. It comprises 25,770 ECG recordings collected from 24,666 patients and includes demographic information such as age and sex. The dataset is structured hierarchically into four classification tasks of increasing granularity: normal versus abnormal ECG classification, ECG category classification, primary statement classification, and full statement classification, with 2, 11, 42, and 117 classes, respectively. To align with our model's hierarchical structure, we focus on the first three classification tasks in a multi-task learning framework, which we designate as task_super, task_sub, and task_diag. The ECG recordings are sampled at 500 Hz with durations ranging from 10 to 60 s, though the majority are 10 s long. Therefore, we retain only 10-s recordings and apply downsampling as necessary. As the dataset is relatively clean, we apply only normalization, without additional preprocessing steps. Table 1 presents the number of ECG recordings in different splits of two datasets, as well as the number of categories for each task, and the total number of ECG recordings.

4.2. Compared models

To comprehensively assess its advantages, we selected CNN-based, RNN-based and Transformer-based time series classification models as baselines.

Table 2

The comparative results of the proposed model with baselines for time series classification based on CNN, RNN, and Transformer on the PTB-XL dataset. Bold numbers indicate the best performance. Underscore indicates the second best performance.

Method		task_super				task_sub				task_diag			
		AUC	F ₁	mAP	Recall	AUC	F ₁	mAP	Recall	AUC	F ₁	mAP	Recall
CNN	FCN_wang [29]	90.43	76.79	87.67	83.22	84.66	60.60	76.17	62.93	79.91	58.18	73.58	59.40
	ResNet1D_wang [29]	90.78	76.60	87.28	83.18	85.58	61.57	76.08	63.61	80.43	58.35	73.16	58.91
	TCN [12]	<u>91.28</u>	76.99	87.95	84.33	85.47	<u>63.49</u>	<u>76.73</u>	<u>68.22</u>	<u>83.62</u>	<u>59.94</u>	73.78	<u>62.61</u>
	MobileNetV3 [30]	88.23	73.47	85.60	80.30	83.08	59.15	72.17	60.78	79.57	56.13	69.80	56.48
	ATI_CNN [31]	88.98	74.74	85.97	77.97	82.76	56.56	71.07	58.03	79.53	54.02	68.32	54.73
	InceptionTime [32]	91.12	78.03	88.38	84.16	82.95	59.80	74.97	61.49	76.63	55.73	71.78	56.50
	DSN [33]	89.97	75.88	87.19	83.79	83.55	57.45	73.86	59.35	77.42	53.84	70.11	54.55
	Scinet [34]	53.23	44.42	67.99	53.68	52.86	41.05	52.84	40.97	48.64	40.99	47.96	40.94
	TimesNet [35]	77.42	60.08	77.83	68.72	75.13	50.66	66.36	54.18	69.88	49.55	64.34	52.93
	ModernTCN [36]	82.95	66.86	82.07	74.55	76.60	53.71	68.61	58.57	69.28	52.56	66.66	56.93
RNN	LSTM [37]	61.83	41.03	65.67	40.95	48.66	41.02	56.50	40.94	48.46	41.02	54.93	40.94
	GRU [38]	62.34	41.03	65.46	40.94	50.25	41.14	55.84	40.94	46.05	41.02	54.76	40.94
	BiLSTM [39]	65.74	51.34	72.02	53.95	61.65	44.31	60.59	43.59	59.59	43.64	58.56	43.02
Transformer	Transformer [27]	78.34	61.02	76.34	73.17	77.17	51.04	66.41	58.22	74.30	50.17	63.70	55.78
	Reformer [40]	75.52	58.29	74.55	67.49	74.88	49.66	65.01	54.65	70.87	48.82	62.67	53.53
	Informer [41]	76.62	59.22	76.33	77.12	73.91	51.80	65.66	57.56	71.47	50.58	63.53	55.63
	ViT [42]	88.89	73.39	85.23	80.03	83.78	60.11	73.27	62.83	79.21	57.08	70.43	58.54
	TNT [43]	89.58	74.78	86.36	82.41	84.62	60.69	73.63	62.99	79.13	56.76	70.23	57.34
	Autoformer [44]	82.12	61.43	77.49	70.56	73.40	48.88	64.61	55.61	69.28	47.35	61.83	54.16
	Crossformer [45]	74.44	57.96	74.47	69.99	75.54	50.49	64.69	51.68	75.08	49.79	63.16	50.76
	FEDformer [46]	67.41	44.90	64.05	53.63	60.35	41.02	55.01	40.94	54.71	41.02	53.41	40.94
	PatchTST [47]	85.09	68.69	83.12	76.34	81.04	55.54	70.52	60.32	77.08	54.02	68.32	57.65
	iTransformer [48]	73.11	57.12	74.83	61.29	74.48	47.40	64.71	47.35	73.11	57.12	<u>74.83</u>	46.50
Ours		91.30	<u>77.56</u>	88.45	83.22	88.02	64.61	78.01	70.03	85.05	62.28	74.96	66.76

Table 3

The comparative results of the proposed model with baselines for time series classification based on CNN, RNN, and Transformer on the SPH dataset. Bold numbers indicate the best performance. Underscore indicates the second best performance.

Method		task_super				task_sub				task_diag			
		AUC	F ₁	mAP	Recall	AUC	F ₁	mAP	Recall	AUC	F ₁	mAP	Recall
CNN	FCN_wang [29]	79.79	77.33	87.29	90.77	80.23	68.26	80.89	69.81	69.78	61.42	74.42	61.25
	ResNet1D_wang [29]	<u>88.25</u>	<u>83.86</u>	<u>91.33</u>	88.86	83.15	<u>77.45</u>	<u>86.47</u>	<u>79.64</u>	63.83	65.97	78.73	65.32
	TCN [12]	82.28	78.50	88.67	90.88	<u>84.86</u>	71.13	83.15	74.01	<u>73.41</u>	65.50	77.46	67.75
	MobileNetV3 [30]	87.01	82.74	90.66	87.64	82.79	74.95	84.91	77.03	72.24	66.78	79.39	66.88
	ATI_CNN [31]	87.44	83.43	91.24	89.76	74.45	68.92	79.39	71.47	67.05	61.70	73.87	61.16
	InceptionTime [32]	85.34	81.06	89.44	90.03	81.54	69.80	82.31	71.57	64.42	61.74	76.29	62.23
	DSN [33]	77.02	74.48	86.10	87.80	73.79	63.91	77.99	64.72	52.22	60.12	72.32	60.06
	Scinet [34]	57.63	66.67	81.33	90.23	51.93	59.09	74.63	59.12	51.50	59.10	69.32	59.10
	TimesNet [35]	62.64	68.31	82.12	83.55	71.72	61.95	76.75	62.74	66.71	60.95	72.45	61.20
	ModernTCN [36]	84.26	80.60	89.26	87.59	79.83	72.73	83.91	77.71	69.36	66.37	79.21	<u>70.26</u>
RNN	LSTM [37]	52.98	66.67	79.55	59.10	48.09	59.10	74.47	59.10	50.26	59.10	65.79	59.10
	GRU [38]	56.09	66.68	79.58	91.11	51.29	59.13	74.53	59.12	50.78	59.10	69.37	59.12
	BiLSTM [39]	55.79	66.67	79.56	60.01	50.92	60.10	74.54	60.01	52.40	60.01	69.72	60.10
Transformer	Transformer [27]	62.39	66.58	81.83	74.16	70.57	61.71	76.16	63.49	66.86	60.99	72.30	62.33
	Reformer [40]	61.17	66.05	81.91	75.12	70.20	53.29	69.43	65.52	66.12	53.60	67.46	64.67
	Informer [41]	59.05	67.00	76.28	61.65	71.51	59.53	76.76	60.30	65.33	58.92	70.42	59.70
	ViT [42]	70.67	70.08	84.72	91.03	62.56	61.64	76.47	63.32	57.45	59.13	69.99	59.14
	TNT [43]	59.78	67.18	81.27	<u>92.11</u>	58.10	60.71	75.90	60.76	48.15	59.10	69.53	59.11
	Autoformer [44]	67.15	68.45	82.76	77.89	67.15	68.46	69.71	56.24	58.63	41.81	62.05	61.09
	Crossformer [45]	59.19	65.67	80.69	79.47	69.72	61.35	76.41	62.56	65.26	60.55	71.71	61.40
	FEDformer [46]	59.42	65.75	72.92	94.91	64.33	33.68	36.49	78.17	59.60	23.55	27.86	64.20
	PatchTST [47]	70.24	70.77	83.82	86.21	72.46	62.38	72.59	62.38	65.52	57.91	72.59	60.83
	iTransformer [48]	55.34	66.67	79.66	59.10	51.59	59.10	74.42	59.10	51.58	59.10	67.96	59.10
Ours		91.62	86.40	92.68	90.08	93.45	80.67	89.19	85.41	82.18	75.09	84.80	79.04

4.2.1. CNN-based model

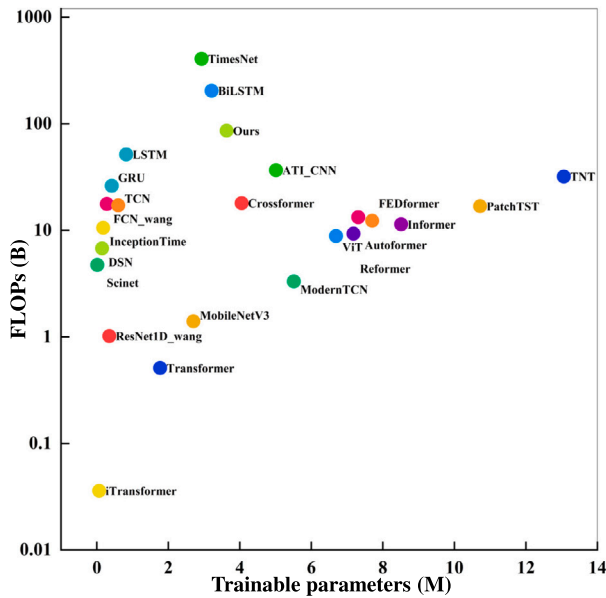
FCN_wang [29]: This model is a complete CNN consisting of 3 convolutional layers and a global average pooling layer. ResNet1d_wang [29]: This model is composed of 3 residual blocks and a global average pooling layer. By introducing residual blocks, it addresses the degradation problem. TCN [12]: This model consists of dilated, causal 1D convolutional layers with matching input and output lengths. MobileNetV3 [30]: MobileNetV3 is an efficient and lightweight neural network that combines depthwise separable convolutions and linear bottlenecks to achieve high accuracy. ATI_CNN [31]: ATI_CNN is a model that integrates CNN, recurrent unit and attention mechanism.

InceptionTime [32]: InceptionTime is a deep learning network using Inception modules and residual connections. DSN [33]: DSN proposes a dynamic sparse network method that automatically adjusts the receptive field size through sparse connections and dynamic sparse training. Scinet [34]: Scinet is a recursive downsampling-convolution-interaction architecture that can efficiently extract multi-scale temporal features. TimesNet [35]: TimesNet is composed of stacked TimesBlocks, which decompose complex temporal variations into different cycles using a modular structure. ModernTCN [36]: ModernTCN uses large convolutional kernels to capture a large receptive field and decouples modeling across three dimensions: time, channels, and variables.

Table 4

Ablation results of LLM module, the router and the hierarchical architecture. Bold numbers indicate the best performance.

Dataset	Ablation	task_super				task_sub				task_diag			
		AUC	F ₁	mAP	Recall	AUC	F ₁	mAP	Recall	AUC	F ₁	mAP	Recall
PTB-XL	w/o LLM	90.36	76.59	87.89	82.72	87.89	63.69	77.79	68.01	84.80	61.93	74.71	66.19
	w/o router	90.93	76.20	87.99	82.14	88.52	63.53	78.05	67.51	84.80	62.01	75.53	65.06
	w/o hierarchical structure	90.32	75.81	87.19	82.44	85.12	58.80	74.20	60.75	77.48	59.04	70.60	57.51
	Ours	91.30	77.56	88.45	83.22	88.02	64.61	78.01	70.03	85.05	62.28	74.96	66.76
SPH	w/o LLM	91.61	85.64	92.71	92.20	93.22	79.81	88.74	84.23	78.12	72.50	83.03	75.35
	w/o router	90.98	85.61	91.89	91.08	92.73	80.54	89.37	84.78	80.81	74.62	84.56	77.41
	w/o hierarchical structure	91.14	85.79	93.21	92.26	92.55	80.19	89.12	85.31	80.85	74.70	84.03	78.64
	Ours	91.62	86.40	92.68	90.08	93.45	80.67	89.19	85.41	82.18	75.09	84.80	79.04

**Fig. 9.** The trainable parameters and FLOPs of the proposed model and baselines.

4.2.2. RNN-based model

LSTM [37]: LSTM addresses the vanishing gradient problem in traditional RNNs by introducing gating mechanisms, enabling it to effectively capture long-term dependencies. GRU [38]: GRU is an efficient RNN structure that simplifies LSTM through a gating mechanism and is suitable for processing sequence data. BiLSTM [39]: BiLSTM extends LSTM by processing sequential data in both forward and backward directions simultaneously, enhancing its ability to capture contextual information.

4.2.3. Transformer-based model

Transformer [27]: Transformer is a model based on the attention mechanism, consisting of an encoder and a decoder. Reformer [40]: Reformer replaces self-attention with locality-sensitive hashing attention, reducing the time complexity. Informer [41]: Informer builds upon Transformer by adding the ProbSparse self-attention mechanism, reducing time complexity. ViT [42]: ViT is a model based on the Transformer architecture for efficient image classification tasks by dividing images into blocks and processing them as sequences. TNT [43]: TNT is a nested Transformer architecture that embeds a local Transformer in a global Transformer. Autoformer [44]: Autoformer discards the self-attention mechanism in Transformer and innovatively introduces an autocorrelation mechanism based on the periodicity of time series. Crossformer [45]: Crossformer uses a two-stage attention mechanism across the time and variable dimensions. FEDformer [46]: FEDformer introduces a module based on Fourier/Wavelet transforms, applying fixed random sampling in the frequency domain to achieve linear complexity

while improving accuracy. PatchTST [47]: PatchTST employs patching and channel independence strategies, achieving excellent results across various time-series problems. iTransformer [48]: iTransformer treats independent sequences as variable tokens, capturing multivariate correlations through the attention mechanism.

4.3. Evaluation metrics

In the experiments, we use the area under the curve (AUC), F₁ score, mean average precision (mAP) and Recall as evaluation metrics. Since the PTB-XL is a multi-label dataset, we use mAP instead of accuracy. The mAP takes into account the precision and Recall values for each label and computes the average across all labels. This metric provides a more comprehensive evaluation of the network's performance in a multi-label classification task. In addition, Recall represents the proportion of all positive samples that are predicted correctly, which has practical implications for disease classification.

4.4. Implementation details

The PyTorch framework is used to achieve the models in the experiments. We use the AdamW optimizer to optimize the model during training, and the learning rate is 0.0001. The total number of training epochs is 100, and the batch size is 64. Our experiments are performed on a single RTX 2080-Ti GPU.

5. Experimental result

5.1. Comparative experiments in multi-task classification

The model performs a hierarchical multi-task classification task, which involves task_super, task_sub, and task_diag. Tables 2 and 3 present the comparison of four evaluation metrics among different models on the PTB-XL dataset and SPH dataset, respectively. Overall, our proposed model achieves the best performance in almost all evaluation metrics of the three tasks in both datasets. On the PTB-XL dataset, for task_super, while the F₁ score of InceptionTime and the Recall of TCN slightly surpass our model, our model achieves the best results for all other metrics. In task_sub, our model demonstrates over a 1% improvement across four evaluation metrics. For task_diag, the F₁ score and Recall of our proposed model reach 62.28% and 66.76%, respectively, outperforming other methods by over 2% and 4%. On the SPH dataset, our model achieved optimal performance on almost all tasks and evaluation metrics. These results indicate that our model achieves superior performance in hierarchical classification tasks, highlighting the advantages of our multi-scale hierarchical structure and its effectiveness in detecting real diseases for ECG signal.

The inferior performance of CNN, RNN, and Transformer-based baselines compared to our model stems from their architectural limitations. RNNs struggle with long-range dependencies and computational inefficiency, while Transformers require large datasets and lack inductive biases tailored for ECG signals. CNNs, though effective in capturing local spatial features, may not fully leverage multi-scale

Table 5

The comparative results of our model (multi-task), our model(single-task), and the baselines (single-task) on the PTB-XL dataset. Bold numbers indicate the best performance. Underscore indicates the second best performance.

Method		task_super				task_sub				task_diag			
		AUC	F ₁	mAP	Recall	AUC	F ₁	mAP	Recall	AUC	F ₁	mAP	Recall
CNN	FCN_wang(single-task) [29]	90.11	76.10	87.55	<u>83.09</u>	81.39	64.02	77.26	66.04	77.81	50.36	68.26	49.75
	ResNet1D_wang(single-task) [29]	<u>91.07</u>	<u>77.46</u>	87.56	82.76	86.97	65.30	77.67	67.25	<u>84.84</u>	<u>62.24</u>	74.41	64.94
	TCN(single-task) [12]	90.79	76.30	87.55	82.45	84.90	63.60	76.91	68.52	84.49	61.16	73.82	65.63
	MobileNetV3(single-task) [30]	87.97	74.15	85.56	80.67	86.04	61.37	74.65	63.71	83.98	58.19	71.39	58.81
	ATL_CNN(single-task) [31]	88.79	74.10	85.47	78.22	86.69	63.02	75.77	65.08	84.29	60.82	73.46	61.65
	InceptionTime(single-task) [32]	90.82	76.99	<u>87.71</u>	82.91	88.75	<u>65.40</u>	79.34	68.89	82.87	61.68	<u>74.81</u>	63.57
	DSN(single-task) [33]	89.50	74.97	86.89	82.08	86.30	61.73	75.85	64.46	78.64	59.50	73.64	61.26
	Scinet(single-task) [34]	71.62	56.57	74.38	64.56	50.56	40.90	52.09	41.04	47.53	40.38	46.88	40.46
	TimesNet(single-task) [35]	76.87	60.72	77.64	70.00	74.28	49.17	64.06	50.33	68.19	50.55	64.33	55.91
	ModernTCN(single-task) [36]	82.97	66.84	82.00	74.64	76.05	52.30	68.18	56.41	67.79	51.10	64.75	53.90
RNN	LSTM(single-task) [37]	74.44	57.90	75.76	64.61	45.59	41.02	56.55	40.94	44.47	41.02	54.86	40.94
	GRU(single-task) [38]	87.37	71.54	84.33	77.24	51.23	41.02	56.56	40.94	46.70	41.02	54.98	40.94
	BiLSTM(single-task) [39]	82.89	65.57	80.04	72.08	47.89	41.02	56.57	40.94	44.47	41.02	55.13	40.94
Transformer	Transformer(single-task) [27]	77.37	60.51	76.96	70.83	72.09	48.20	63.00	51.74	73.39	49.48	62.82	54.49
	Reformer(single-task) [40]	75.14	56.60	74.11	61.13	74.86	47.35	62.87	55.21	72.32	48.14	61.24	54.81
	Informer(single-task) [41]	76.41	59.40	76.29	74.72	75.52	50.35	65.04	55.61	71.88	45.63	60.08	51.16
	ViT(single-task) [42]	88.31	73.63	85.48	78.47	85.61	61.79	74.65	64.12	83.66	59.14	72.60	60.31
	TNT(single-task) [43]	90.11	75.87	86.95	79.94	86.40	62.21	75.61	65.01	84.62	61.87	74.62	63.70
	Autoformer(single-task) [44]	81.33	51.64	75.63	65.09	70.60	32.27	48.55	45.93	68.38	44.60	59.05	52.18
	Crossformer(single-task) [45]	71.48	55.45	74.12	63.87	75.39	47.44	64.47	47.96	71.60	46.89	61.34	47.58
	FEDformer(single-task) [46]	74.68	54.41	72.97	64.28	69.44	29.05	54.87	59.34	57.79	28.00	44.59	55.71
	PatchTST(single-task) [47]	83.75	67.45	81.71	75.31	79.09	55.23	69.86	61.76	76.85	54.35	68.69	57.75
	iTransformer(single-task) [48]	74.02	56.61	75.27	60.99	72.74	45.69	63.42	46.02	72.31	46.34	62.58	46.25
Ours (single-task)		90.40	75.65	87.47	81.97	<u>88.21</u>	66.18	77.93	71.60	84.28	62.09	74.25	<u>66.51</u>
Ours (multi-task)		91.30	77.56	88.45	83.22	88.02	64.61	<u>78.01</u>	<u>70.03</u>	85.05	62.28	74.96	66.76

temporal dependencies. Among the baselines, TCN, InceptionTime, and ResNet1D_wang perform better due to their advanced temporal modeling capabilities. TCN efficiently captures long-range dependencies with dilated convolutions, InceptionTime leverages multi-scale convolutional kernels for robust feature extraction, and ResNet1D_wang benefits from deep residual connections that enhance gradient flow and representation learning. Overall, CNN-based baselines outperform RNNs and Transformers in ECG classification, as they better preserve spatial and temporal features while maintaining computational efficiency. However, their lack of hierarchical and adaptive mechanisms limits their performance compared to our model, which integrates multi-scale embeddings, hierarchical routing, and LLM-driven contextual learning.

In addition, to evaluate the computational efficiency of our proposed model, we conduct experiments analyzing the number of trainable parameters and FLOPs of our proposed model and baselines on the PTB-XL dataset. As presented in Fig. 9, since the pre-trained LLM is frozen, our model maintains a moderate number of trainable parameters, avoiding excessive computational overhead. To enhance performance, our model integrates a dual-branch architecture combining Transformer and TCN, incorporating a multi-scale strategy and leveraging the LLM. As a result, its computational cost is slightly higher than that of single-branch, single-scale networks that do not utilize a LLM.

5.2. Ablation studies

We conduct focused ablation studies on various modules of our model, including the LLM module, the router, and the hierarchical architecture, to evaluate their contributions. For the hierarchical architecture ablation, we merge all scale outputs from each branch into one final vector, and then combine the vectors of two branches into one final vector, which is subsequently fed into the three classification heads. The ablation results are shown in Table 4. Removing the LLM module results in a significant decline in classification performance across all three tasks. This demonstrates the importance of leveraging the LLM to integrate static information with time series data. By converting static information into prompt vectors, the LLM provides contextual awareness, enhancing model generalization. In addition, removing the router leads to a notable performance drop, emphasizing its critical

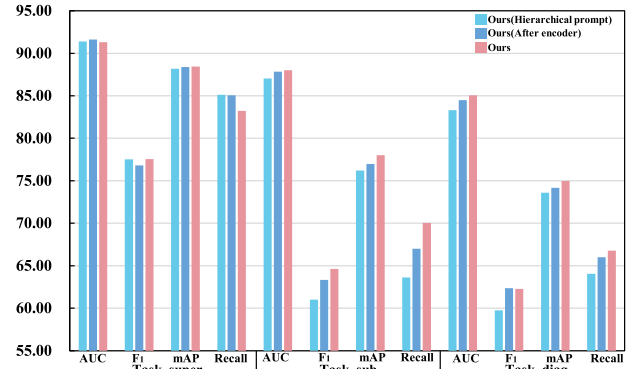


Fig. 10. Performance of different LLM strategies in three tasks on the PTB-XL dataset.

role. The router not only aggregates information from individual scales but also establishes cross-scale correlations. This enables the model to integrate multi-scale information, yielding a more comprehensive feature representation that significantly improves performance. Finally, the hierarchical architecture applies features at different scales to corresponding classification tasks and uses the outputs from previous tasks as contextual information for subsequent tasks. This hierarchical multi-task classification strategy refines the classification process by embedding multi-scale features, transmitting information progressively, and enhancing task granularity, thereby substantially boosting model accuracy.

5.3. Comparative experiments in single-task classification

Our model adopts a multi-task architecture, enabling it to simultaneously perform three classification tasks. This multi-task design reduces computational overhead, making the model more efficient. To further demonstrate the effectiveness of our model, we modify both

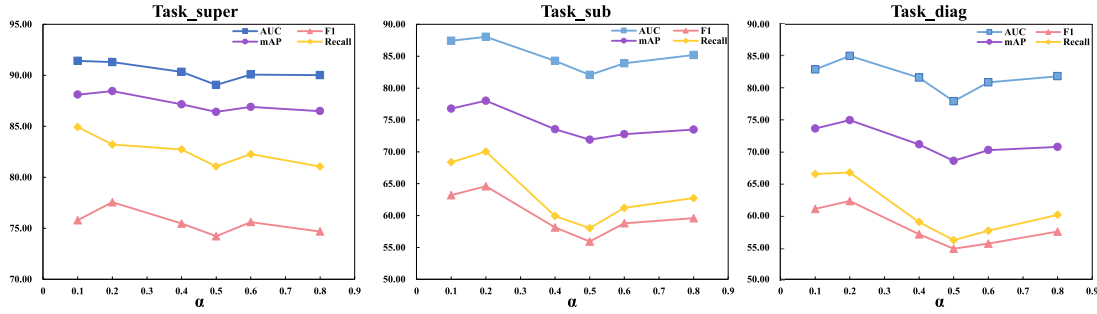


Fig. 11. Performance of different fusion parameters in three tasks on the PTB-XL dataset.

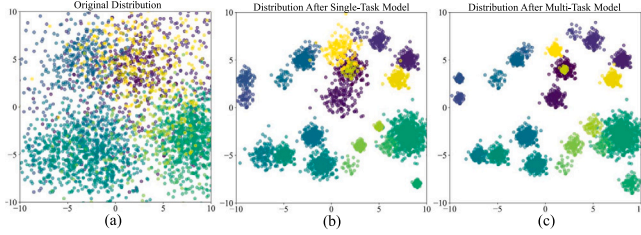


Fig. 12. The t-SNE visualization results. (a) Original distribution of 23 categories in the task_sub. (b) Distribution of 23 categories in the task_sub after processing with our single-task model. (c) Distribution of 23 categories in the task_sub after processing with our multi-task model.

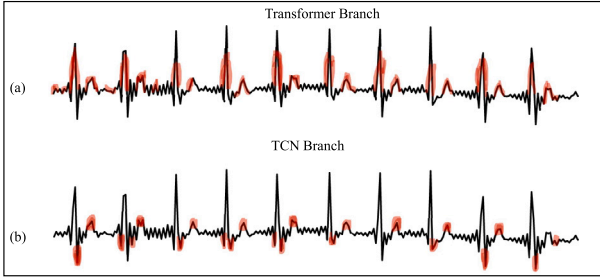


Fig. 13. Visualization of ECG features based on the two branches in our proposed network.

our model and the baselines into single-task classification architectures, where each model independently performs the three classification tasks. We compare the performance of our multi-task hierarchical model with these single-task models on the PTB-XL dataset. The comparison results are presented in Table 5. In simple multi-task frameworks, which share a backbone and output through independent classification heads without considering task relationships, performing each classification task separately often outperforms doing them simultaneously. However, in our model, task relationships are explicitly considered, and multi-scale strategies are applied to different levels of classification tasks. As a result, our multi-task hierarchical structure not only outperforms the single-task variant of our model but also surpasses the single-task structures of baseline models. Our multi-task hierarchical model achieves the best results in most cases, except for individual evaluation metrics of individual tasks where our model is slightly lower than the single-task model. This further validates the effectiveness of the proposed model.

5.4. Effect of the patch lengths in the Transformer branch

In this part, we compare the performance of different numbers and configurations of patch lengths, including multiple identical patch lengths, the top 15 periodic lengths (top 15), the top 12 periodic lengths

(top 12), and others. In our experiments, for the PTB-XL dataset, S is set to 15, resulting in a patch length list of [245,157,122,91,80,70,63,59,55,50,49,42,44,43,43]. These values are sorted in descending order and evenly divided into three groups. From each group, we select D patch lengths, where D is set to 3. This selection process yielded a final patch length list for our model: [122,91,80,70,63,59,49,44,43]. The results, summarized in Table 6, demonstrate that patch lengths determine via FFT-based periodicity analysis outperform conventional powers-of-two patch lengths in terms of model effectiveness. This underscores the efficacy of our FFT-based strategy for determining patch lengths. Moreover, using the top 15 or top 12 periodic lengths as patch lengths can surpass our model in certain metrics for specific tasks. However, our model employs only top 9 patch lengths, resulting in significantly reduced computational overhead compared to using top 15 or top 12 periodic lengths. Overall, our patch length selection strategy achieves optimal performance while maintaining computational efficiency.

To validate the superiority of the multi-scale embedding approach over traditional single-scale approaches, we compared the performance of single-scale patch embedding. As shown in Table 7, our multi-scale strategy outperformed the single-scale patch embedding approach across nearly all tasks and evaluation metrics. Furthermore, by comparing Tables 6 and 7, specifically for patch lengths of 128 and 64, we observe that using repeated patch lengths generally yielded better performance on certain metrics than relying on a single patch length. These experimental results confirm that our multi-scale strategy effectively enhances model performance.

5.5. Effect of multi-scale convolution kernels in the TCN branch

We evaluate the impact of varying the size and number of multi-scale convolutional kernels in the TCN branch on classification performance. In the TCN branch of our model, a kernel list of [7,5,3] is selected based on the results of our systematic experimental exploration on the PTB-XL dataset. The experimental results are shown in Table 8. The chosen kernel sizes in our model outperform all other kernel configurations across nearly all tasks and metrics. Except for the F_1 and Recall metrics of task_sub and the AUC metric of task_daig, [19,17,15,13,11,9,7,5,3] outperform our model. However, setting the number of kernels to 9 significantly increases computational complexity compared to our setting. This observation suggests that a larger number of kernels does not necessarily lead to better performance. In fact, excessive convolutional kernels do not provide a substantial improvement in model performance and result in a higher computational cost. This also implies that not all scales of convolutional kernels contribute equally to improving classification performance.

5.6. Exploration of LLM strategies

To gain a deeper understanding of the potential and impact of LLM in ECG classification tasks, we explore different LLM strategies on the PTB-XL dataset and analyze their effects on model performance. We

Table 6

The comparative results of different patch lengths in the Transformer branch on the PTB-XL dataset. Bold numbers indicate the best performance.

Patch length	task_super				task_sub				task_diag			
	AUC	F ₁	mAP	Recall	AUC	F ₁	mAP	Recall	AUC	F ₁	mAP	Recall
[128,128,128,128,128,128,128,128]	91.16	76.49	88.35	83.15	87.20	64.21	77.42	69.45	84.15	62.20	74.76	66.35
[64,64,64,64,64,64,64,64]	91.01	76.25	87.96	81.83	88.20	64.27	76.96	69.79	85.05	61.71	74.65	66.37
[128,128,128,64,64,64,32,32]	91.11	77.23	88.18	83.47	88.22	63.56	77.85	66.99	84.48	61.84	73.97	65.71
[64,64,64,32,32,32,16,16]	91.28	76.84	87.89	83.11	86.90	62.61	75.40	66.53	80.32	59.39	72.48	62.10
[256,256,128,128,64,64,32,16,8]	91.24	77.37	87.95	83.18	86.26	63.82	76.64	67.84	84.47	60.22	74.14	63.29
Top 15	90.97	76.63	87.75	83.26	88.27	64.29	77.42	68.61	84.41	62.12	74.84	66.33
Top 12	91.39	76.76	88.15	82.65	87.13	63.36	76.93	68.69	83.32	61.33	73.77	64.28
Ours	91.30	77.56	88.45	83.22	88.02	64.61	78.01	70.03	85.05	62.28	74.96	66.76

Table 7

The comparative results of our multi-scale patch strategy and single-scale patch strategies in the Transformer branch on the PTB-XL dataset. Bold numbers indicate the best performance.

Patch length	task_super				task_sub				task_diag			
	AUC	F ₁	mAP	Recall	AUC	F ₁	mAP	Recall	AUC	F ₁	mAP	Recall
[256]	90.93	76.67	88.04	84.15	86.48	60.57	74.63	63.98	83.34	58.44	71.43	61.10
[128]	91.14	77.10	88.44	85.06	87.80	64.19	77.64	69.41	84.53	61.53	74.49	65.20
[64]	91.01	77.50	88.31	84.57	86.84	64.38	76.94	68.79	83.99	61.75	73.94	65.63
[32]	91.05	77.21	88.32	83.12	88.13	63.51	77.19	68.62	84.07	61.84	74.81	65.48
[16]	91.13	76.97	88.07	83.13	87.78	64.40	77.28	69.63	84.69	62.15	73.98	66.54
Ours	91.30	77.56	88.45	83.22	88.02	64.61	78.01	70.03	85.05	62.28	74.96	66.76

Table 8

The comparative results of multi-scale convolution kernels in the TCN branch on the PTB-XL dataset. Bold numbers indicate the best performance.

Convolution kernels	task_super				task_sub				task_diag			
	AUC	F ₁	mAP	Recall	AUC	F ₁	mAP	Recall	AUC	F ₁	mAP	Recall
[13,7,3]	90.67	75.62	87.27	81.74	87.30	63.00	76.35	67.15	84.33	61.01	74.01	64.23
[13,11,9,7,5,3]	90.81	75.94	87.48	80.38	86.68	62.93	76.30	66.77	85.29	60.41	73.71	63.09
[19,17,15,13,11,9,7,5,3]	90.51	74.33	87.19	81.43	87.88	65.42	78.00	71.32	86.67	61.53	74.89	66.45
Ours	91.30	77.56	88.45	83.22	88.02	64.61	78.01	70.03	85.05	62.28	74.96	66.76

conduct two explorations: designing hierarchical prompt and modifying the fusion position of the prompt embeddings.

The performance of these modifications across three tasks is illustrated in Fig. 10. For the hierarchical prompt design, we modify the prompt template to: “This is a {class}-level ECG classification task with {number} types. This patient is {age} and {gender}”. Compared to the original prompt, this version introduces hierarchical task information and the total number of ECG classes at each level. However, our results indicate that hierarchical prompts performed worse than the original single-prompt template on most evaluation metrics. This suggests that the most influential factor is the inclusion of static patient information, while additional complexity does not contribute to improved model performance.

Regarding the exploration of prompt embedding fusion, we modify the integration approach by merging prompt embeddings with the encoder’s output vectors. In this method, after the ECG classification model independently extracts features, the LLM functions as an auxiliary decision-making module to optimize the final classification. However, experimental results show that this method does not enhance model performance. In our model, prompt embeddings are originally integrated into a multi-scale embedding module and fused with patch embeddings from the ECG data. These findings suggest that incorporating prompt embeddings at the embedding layer provides more effective contextual support, enabling the model to reason more effectively.

5.7. Hyperparameter experiments

For the output vectors from two branches, we use a simple weighted method for fusion. Let α and $(1-\alpha)$ represent the weights assigned to the Transformer and TCN branches, respectively. Based on our systematic experimental exploration, we set $\alpha = 0.2$ in our model. Fig. 11 presents the results of our model for different values of α on

the PTB-XL dataset. If α is set too large or too small, the model does not achieve optimal performance. The Transformer branch excels at extracting global information, while the TCN branch has an advantage in capturing local features. This indicates that the model needs to strike a balance between global and local information, enabling effective complementarity between the two feature types. Such a balance allows for a more comprehensive representation of the temporal characteristics of the time series, ultimately leading to improved model performance.

6. Visualization

In this paper, two visualization methods are used to evaluate our proposed model. First, we apply t-SNE to map high-dimensional data into a lower-dimensional space. Fig. 12(a) shows the raw data distribution from the 23 categories in the task_sub, where significant category overlap indicates weak inherent correlations. Fig. 12(b) presents results from a single-task classification model, revealing better clustering but persistent boundary ambiguity for certain categories. In contrast, Fig. 12(c) illustrates our multi-task hierarchical classification model, where previously unclear boundaries become more distinct. By leveraging task correlations and progressive information flow, our model enhances reasoning and classification performance. The visualization clearly demonstrates the effectiveness of our hierarchical approach. Second, we use Gradient-weighted Class Activation Mapping [49] to highlight important regions influencing model predictions. To illustrate feature extraction of the Transformer branch and the TCN branch, we visualize the features from one lead of the ECG in the dataset. Fig. 13(a) visualizes features from the Transformer branch, showing a focus on global information, while Fig. 13(b) highlights the TCN branch’s emphasis on local information. Our model combines these complementary strengths to improve prediction accuracy by integrating both global and local feature representations.

7. Limitations

Although our model has made significant progress in ECG classification tasks, there are still some limitations. Firstly, our model employs a simple weighted fusion strategy for combining the output vectors from the two branches, which may not fully optimize its performance. Future research could explore more advanced fusion techniques to enhance model effectiveness. Additionally, our approach integrates prompt embeddings with multi-scale patch embeddings, followed by reasoning through an encoder module, which increases computational overhead. Although we investigated different fusion positions and prompt designs in our experiments, none outperformed the original model. Therefore, developing a more efficient fusion strategy that simultaneously reduces computational complexity and improves performance could significantly enhance the overall efficiency of the model.

8. Conclusion

In this paper, to accommodate the hierarchical nature of heart diseases in ECG dataset, we propose a LLM-assisted multi-scale hierarchical ECG signal classification model composed of two branches: Transformer and TCN. The Transformer branch utilizes FFT to extract the periodic characteristics of the signals, with the periodic lengths chosen as multi-scale patch sizes to generate multi-scale embedding vectors. Moreover, static patient information is embedded using a LLM for improved feature representation. Additionally, we introduce a router mechanism to learn intra-scale and inter-scale feature relationships using self-attention. In the TCN branch, convolutional kernels of varying sizes are employed to extract features across different scales, tailored to classification tasks at distinct levels. The feature vectors extracted from each level of the multi-scale Transformer and multi-scale TCN branches are integrated, allowing the model to effectively capture both global and local information. Experimental results demonstrate that our model achieves superior performance compared to existing approaches. Future research can explore more advanced fusion strategies to reduce computational complexity while enhancing model performance, thereby improving overall efficiency.

CRedit authorship contribution statement

Qianjiang Chen: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology. **Cheng Lian:** Writing – review & editing, Writing – original draft, Supervision, Software. **Bingrong Xu:** Supervision, Investigation. **Quan Zhou:** Supervision, Investigation, Conceptualization. **Yixin Su:** Investigation, Funding acquisition, Conceptualization. **Zhigang Zeng:** Supervision, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China under Grants 62176193, 62206204 and 62401414.

Data availability

Data will be made available on request.

References

- [1] S. Yang, C. Lian, Z. Zeng, B. Xu, J. Zang, Z. Zhang, A multi-view multi-scale neural network for multi-label ECG classification, *IEEE Trans. Emerg. Top. Comput. Intell.* 7 (3) (2023) 648–660.
- [2] M. Zhou, L. Chen, X. Wei, X. Liao, Q. Mao, H. Wang, H. Pu, J. Luo, T. Xiang, B. Fang, Perception-oriented U-shaped transformer network for 360-degree no-reference image quality assessment, *IEEE Trans. Broadcast.* 69 (2) (2023) 396–405.
- [3] M. Zhou, X. Lan, X. Wei, X. Liao, Q. Mao, Y. Li, C. Wu, T. Xiang, B. Fang, An end-to-end blind image quality assessment method using a recurrent network and self-attention, *IEEE Trans. Broadcast.* 69 (2) (2022) 369–377.
- [4] M. Zhou, X. Zhao, F. Luo, J. Luo, H. Pu, T. Xiang, Robust rgb-t tracking via adaptive modality weight correlation filters and cross-modality learning, *ACM Trans. Multimed. Comput. Commun. Appl.* 20 (4) (2023) 1–20.
- [5] M. Zhou, W. Shen, X. Wei, J. Luo, F. Jia, X. Zhuang, W. Jia, Blind image quality assessment: Exploring content fidelity perceptibility via quality adversarial learning, *Int. J. Comput. Vis.* (2025) 1–17.
- [6] M. Zhou, X. Wu, X. Wei, T. Xiang, B. Fang, S. Kwong, Low-light enhancement method based on a retinex model for structure preservation, *IEEE Trans. Multimed.* 26 (2023) 650–662.
- [7] J. Zhang, J. Tian, Y. Cao, Y. Yang, X. Xu, Deep time-frequency representation and progressive decision fusion for ECG classification, *Knowl.-Based Syst.* 190 (2020) 105402.
- [8] P. Schwab, G.C. Scebba, J. Zhang, M. Delai, W. Karlen, Beat by beat: Classifying cardiac arrhythmias with recurrent neural networks, in: 2017 Computing in Cardiology, Cinc, IEEE, 2017, pp. 1–4.
- [9] G. Yan, S. Liang, Y. Zhang, F. Liu, Fusing transformer model with temporal features for ECG heartbeat classification, in: 2019 IEEE International Conference on Bioinformatics and Biomedicine, BIBM, IEEE, 2019, pp. 898–905.
- [10] P. Wagner, N. Strodthoff, R.-D. Boussejot, D. Kreiseler, F.I. Lunze, W. Samek, T. Schaeffter, PTB-XL, a large publicly available electrocardiography dataset, *Sci. Data* 7 (1) (2020) 1–15.
- [11] J.L. Rodgers, J. Jones, S.I. Bolleddu, S. Vanthenapalli, L.E. Rodgers, K. Shah, K. Karia, S.K. Panguluri, Cardiovascular risks associated with gender and aging, *J. Cardiovasc. Dev. Dis.* 6 (2) (2019) 19.
- [12] S. Bai, J.Z. Kolter, V. Koltun, An empirical evaluation of generic convolutional and recurrent networks for sequence modeling, 2018, arXiv preprint arXiv:1803.01271.
- [13] W. Wang, W. Chen, Q. Qiu, L. Chen, B. Wu, B. Lin, X. He, W. Liu, Crossformer++: A versatile vision transformer hinging on cross-scale attention, *IEEE Trans. Pattern Anal. Mach. Intell.* (2023).
- [14] K.D. Tzimourta, V. Christou, A.T. Tzallas, N. Giannakeas, L.G. Astrakas, P. Angelidis, D. Tsalikakis, M.G. Tsipouras, Machine learning algorithms and statistical approaches for Alzheimer's disease analysis based on resting-state EEG recordings: A systematic review, *Int. J. Neural Syst.* 31 (05) (2021) 2130002.
- [15] J.-Z. Jian, T.-R. Ger, H.-H. Lai, C.-M. Ku, C.-A. Chen, P.A.R. Abu, S.-L. Chen, Detection of myocardial infarction using ECG and multi-scale feature concatenate, *Sensors* 21 (5) (2021) 1906.
- [16] R. Wang, J. Fan, Y. Li, Deep multi-scale fusion neural network for multi-class arrhythmia detection, *IEEE J. Biomed. Heal. Inform.* 24 (9) (2020) 2461–2472.
- [17] Ö. Yildirim, P. Plawiak, R.-S. Tan, U.R. Acharya, Arrhythmia detection using deep convolutional neural network with long duration ECG signals, *Comput. Biol. Med.* 102 (2018) 411–420.
- [18] B. Hou, J. Yang, P. Wang, R. Yan, LSTM-based auto-encoder model for ECG arrhythmias classification, *IEEE Trans. Instrum. Meas.* 69 (4) (2019) 1232–1240.
- [19] A. Natarajan, Y. Chang, S. Mariani, A. Rahman, G. Boverman, S. Vij, J. Rubin, A wide and deep transformer neural network for 12-lead ECG classification, in: 2020 Computing in Cardiology, IEEE, 2020, pp. 1–4.
- [20] H. Yan, Z. Li, W. Li, C. Wang, M. Wu, C. Zhang, ConTNet: Why not use convolution and transformer at the same time? 2021, arXiv preprint arXiv:2104.13497.
- [21] X. Wu, Y. Feng, H. Xu, Z. Lin, T. Chen, S. Li, S. Qiu, Q. Liu, Y. Ma, S. Zhang, CTransCNN: Combining transformer and CNN in multilabel medical image classification, *Knowl.-Based Syst.* 281 (2023) 111030.
- [22] X. Zhang, R.R. Chowdhury, R.K. Gupta, J. Shang, Large language models for time series: A survey, 2024, arXiv preprint arXiv:2402.01801.
- [23] M. Jin, S. Wang, L. Ma, Z. Chu, J.Y. Zhang, X. Shi, P.-Y. Chen, Y. Liang, Y.-F. Li, S. Pan, et al., Time-llm: Time series forecasting by reprogramming large language models, 2023, arXiv preprint arXiv:2310.01728.
- [24] H. Xue, F.D. Salim, Promptcast: A new prompt-based learning paradigm for time series forecasting, *IEEE Trans. Knowl. Data Eng.* (2023).
- [25] D. Cao, F. Jia, S.O. Arik, T. Pfister, Y. Zheng, W. Ye, Y. Liu, Tempo: Prompt-based generative pre-trained transformer for time series forecasting, 2023, arXiv preprint arXiv:2310.04948.
- [26] J.D.M.-W.C. Kenton, L.K. Toutanova, Bert: Pre-training of deep bidirectional transformers for language understanding, in: Proceedings of NaacL-HLT, vol. 1, Minneapolis, Minnesota, 2019, p. 2.
- [27] A. Vaswani, Attention is all you need, *Adv. Neural Inf. Process. Syst.* (2017).

- [28] H. Liu, D. Chen, D. Chen, X. Zhang, H. Li, L. Bian, M. Shu, Y. Wang, A large-scale multi-label 12-lead electrocardiogram database with standardized diagnostic statements, *Sci. Data* 9 (1) (2022) 272.
- [29] Z. Wang, W. Yan, T. Oates, Time series classification from scratch with deep neural networks: A strong baseline, in: 2017 International Joint Conference on Neural Networks, IJCNN, IEEE, 2017, pp. 1578–1585.
- [30] A. Howard, M. Sandler, G. Chu, L.-C. Chen, B. Chen, M. Tan, W. Wang, Y. Zhu, R. Pang, V. Vasudevan, et al., Searching for mobilenetv3, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 1314–1324.
- [31] Q. Yao, R. Wang, X. Fan, J. Liu, Y. Li, Multi-class arrhythmia detection from 12-lead varied-length ECG using attention-based time-incremental convolutional neural network, *Inf. Fusion* 53 (2020) 174–182.
- [32] H. Ismail Fawaz, B. Lucas, G. Forestier, C. Pelletier, D.F. Schmidt, J. Weber, G.I. Webb, L. Idoumghar, P.-A. Muller, F. Petitjean, Inceptiontime: Finding alexnet for time series classification, *Data Min. Knowl. Discov.* 34 (6) (2020) 1936–1962.
- [33] Q. Xiao, B. Wu, Y. Zhang, S. Liu, M. Pechenizkiy, E. Mocanu, D.C. Mocanu, Dynamic sparse network for time series classification: Learning what to “see”, *Adv. Neural Inf. Process. Syst.* 35 (2022) 16849–16862.
- [34] M. Liu, A. Zeng, M. Chen, Z. Xu, Q. Lai, L. Ma, Q. Xu, Scinet: Time series modeling and forecasting with sample convolution and interaction, *Adv. Neural Inf. Process. Syst.* 35 (2022) 5816–5828.
- [35] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, M. Long, Timesnet: Temporal 2d-variation modeling for general time series analysis, 2022, arXiv preprint arXiv:2210.02186.
- [36] D. Luo, X. Wang, ModernTCN: A modern pure convolution structure for general time series analysis, in: The Twelfth International Conference on Learning Representations, 2024.
- [37] J. Schmidhuber, S. Hochreiter, et al., Long short-term memory, *Neural Comput.* 9 (8) (1997) 1735–1780.
- [38] K. Cho, B. Van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, Y. Bengio, Learning phrase representations using RNN encoder-decoder for statistical machine translation, 2014, arXiv preprint arXiv:1406.1078.
- [39] S. Zhang, D. Zheng, X. Hu, M. Yang, Bidirectional long short-term memory networks for relation classification, in: Proceedings of the 29th Pacific Asia Conference on Language, Information and Computation, 2015, pp. 73–78.
- [40] N. Kitaev, L. Kaiser, A. Levskaya, Reformer: The efficient transformer, 2020, arXiv preprint arXiv:2001.04451.
- [41] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, W. Zhang, Informer: Beyond efficient transformer for long sequence time-series forecasting, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 35, 2021, pp. 11106–11115, 12.
- [42] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al., An image is worth 16x16 words: Transformers for image recognition at scale, 2020, arXiv preprint arXiv:2010.11929.
- [43] K. Han, A. Xiao, E. Wu, J. Guo, C. Xu, Y. Wang, Transformer in transformer, *Adv. Neural Inf. Process. Syst.* 34 (2021) 15908–15919.
- [44] H. Wu, J. Xu, J. Wang, M. Long, Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting, *Adv. Neural Inf. Process. Syst.* 34 (2021) 22419–22430.
- [45] Y. Zhang, J. Yan, Crossformer: Transformer utilizing cross-dimension dependency for multivariate time series forecasting, in: The Eleventh International Conference on Learning Representations, 2023.
- [46] T. Zhou, Z. Ma, Q. Wen, X. Wang, L. Sun, R. Jin, Fedformer: Frequency enhanced decomposed transformer for long-term series forecasting, in: International Conference on Machine Learning, PMLR, 2022, pp. 27268–27286.
- [47] Y. Nie, N.H. Nguyen, P. Sinthong, J. Kalagnanam, A time series is worth 64 words: Long-term forecasting with transformers, 2022, arXiv preprint arXiv:2211.14730.
- [48] Y. Liu, T. Hu, H. Zhang, H. Wu, S. Wang, L. Ma, M. Long, itransformer: Inverted transformers are effective for time series forecasting, 2023, arXiv preprint arXiv:2310.06625.
- [49] R.R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra, Grad-cam: Visual explanations from deep networks via gradient-based localization, in: Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 618–626.